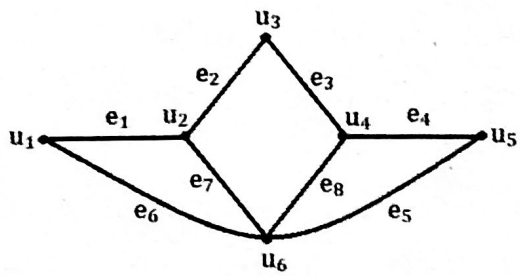




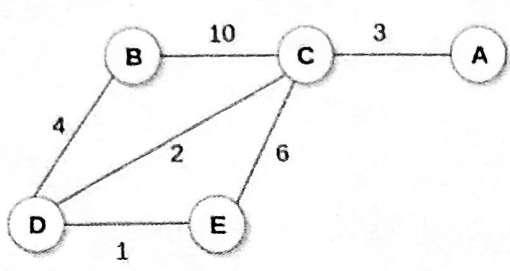
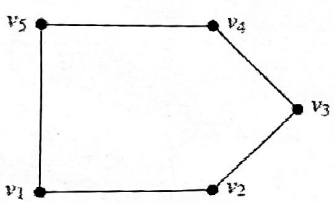
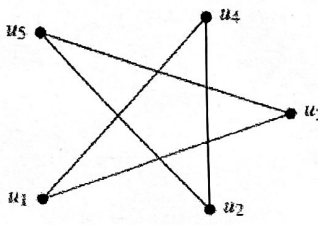
DEPARTMENT OF MATHEMATICS

Sub Code:	AI 34/AD 34/CI 35/CY 35	Sub:	Discrete mathematical Structures	Test:	II
Time:	3.00 pm-4.00 pm.	Term:	20.11.23 to 2.03.24	Marks:	30
Date:	27.02.2024	Semester:	III	Sections:	AI/AD/CI/CY

Note: Answer any TWO full questions. Each main question carries 15 marks.

Q.No.	Questions	CO's	Blooms level	Marks
1.	(a) Solve the recurrence relation $a_n = 7a_{n-1} - 12a_{n-2}$, $a_0 = 3, a_1 = 11$.	CO2	L1	5
	(b) Draw the graph of the following adjacency matrix and hence write its incidence matrix. $A = \begin{bmatrix} & u & v & w & x & y \\ u & 0 & 1 & 1 & 0 & 1 \\ v & 1 & 0 & 1 & 0 & 1 \\ w & 1 & 1 & 0 & 1 & 1 \\ x & 0 & 0 & 1 & 0 & 1 \\ y & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$	CO3	L3	5
	(c) Prove that a tree with n vertices has $n-1$ edges.	CO4	L3	5
2.	(a) Consider $D_{70} = \{1, 2, 5, 7, 10, 14, 35, 70\}$, the divisors of 70 with divisibility as order relation. Verify whether it is a Boolean algebra or not?	CO2	L2	5
	(b) Find (i) Fusion of u_1 and u_2 (ii) $G - e_1$ (iii) $G - u_1$ for the following graph. 	CO3	L4	5

(PTO)

	(c)	Write Prim's algorithm to find the shortest spanning tree and hence find the minimal spanning tree of the following graph:	CO4	L3	5
					
3.	(a)	Find an explicit formula for the sequence $a_n = a_{n-1} + 3^n$ given $a_1 = 4$.	CO2	L4	5
	(b)	Define (i) walk (ii) path in a graph G. (iii) Verify whether the following graphs are isomorphic or not?	CO3	L2	5
		 (a) G_1.  (b) G_2.			
	(c)	Define semi group. Let G be the set of real numbers excluding 1 and $*$ be defined by $a*b = a+b-ab$ for $a, b \in G$ Check whether G is monoid or not?	CO5	L3	5